Methodology Overview

Deliverable D3, Capstone Portfolio
Programme: MSc Data Science & Society, Tilburg University
Student: Noah Tjon Sien Kie (2103939)
Supervisor: Dr. Sharon Ong
Project: Cross-Dataset Generalisation of Deep Learning for Malaria Classification in Microscopy Images
Repository: https://github.com/noahtjonsk/malaria-cross-dataset
Date: October 2026

1. Introduction

This Methodology Overview (MO) documents how the project is implemented and why each methodological choice was made: where the data come from, how the cells are cropped, cleaned and split, which models are trained and compared, and how the results are evaluated. Each step names the script or function that implements it, so anyone with the three public datasets can rerun the work. Together with the literature review and the proof-of-concept code in the repository, it forms the Minimum Viable Product assessed at the Midterm Portfolio Review. It should be read alongside:

- the literature review, `LiteratureReview_NoahTjonSienKie.pdf`;
- the project requirements and run order, `README.md`, with pinned packages in `requirements.txt`;
- the exploratory data analysis, `notebooks/03_eda_short.ipynb` (code-free copy `outputs/03_eda_short.html`), with the full record in `notebooks/01_eda.ipynb`;
- the ELSA report (D5), `docs/elsa_report.md`;
- the preliminary results in §5.6, read from `outputs/tables/rq{1,2,3}_<model>_s0.csv`, `rq1_f1_<model>_s0.csv` and the model comparison `rq1_compare_s0.csv`.

1.1 Motivation, gap and research questions

CNNs that classify single red-cell images as parasitised or uninfected reach near-perfect scores on the NIH dataset, but lose much of that performance on images from other laboratories. The literature review found that no study varies crop format or stain colour on its own while holding the model fixed, so the cause of the cross-dataset drop is still open. Each research question below is a supervised-learning measurement that addresses part of this gap: RQ1 is a model comparison under one shared shift, RQ2 is an ablation on the test inputs, and RQ3 is an error analysis by subgroup.

Three ImageNet-pretrained CNNs (VGG-16, ResNet-50 and MobileNetV2) are fine-tuned on NIH single-cell images of *P. falciparum* from Chittagong (Rajaraman et al., 2018). They are then applied, without fine-tuning, to two external test sets: BBBC041, *P. vivax* from two acquisition batches (Ljosa et al., 2012; Hung & Carpenter, 2017), and MP-IDB, four species from Lausanne (Loddo et al., 2019). The trained models never change after RQ1, so any change in RQ2 can be attributed to the step that caused it.

MRQ. To what extent do CNNs trained to classify NIH single-cell images as parasitised or uninfected lose sensitivity on BBBC041 and MP-IDB, and specificity on BBBC041, relative to a patient-grouped NIH hold-out set, and which dataset differences explain this drop?

RQ1. How large is the drop from the patient-grouped NIH hold-out set, in percentage points, for MobileNetV2, ResNet-50 and VGG-16 applied without fine-tuning to BBBC041 and MP-IDB, and does the

lighter architecture lose less than the heavier ones?

RQ2. What share of the RQ1 drop is recovered when test cells are matched to NIH at test time, first in crop format (background removed) and then in stain colour (Reinhard or histogram matching)?

RQ3. Does sensitivity on the pooled different-species MP-IDB cells (*n* = 140) differ from *P. falciparum* (*n* = 1,297) by more than the resampling interval, and which BBBC041 life stages account for the largest share of missed parasites?

RQ1 is answered in stages E and G, RQ2 in stages F and G, and RQ3 in stage H. Sections 2 and 3 describe the data and how it is prepared (stages A to C), Section 4 the EDA (stage D), Section 5 the models, the RQ2 test inputs and the evaluation (stages E to H), and Section 6 how to reproduce everything.

1.2 Pipeline overview

Figure 1 shows the pipeline grouped by research question. Letters A to H are used throughout this document. Solid boxes are implemented and produced the results in §5.6; dashed boxes are planned for the final results.



Figure 1. Pipeline and study design. Each model is trained once on NIH (E), tested frozen on the raw test crops (G, RQ1), tested again on edited copies of the same crops (F, G, RQ2), and its errors are broken down by species and life stage (H3, RQ3). Drawn by `scripts/make_mo_figures.py`.

2. Dataset description and access (stage A)

2.1 Sources and access

All three datasets are public. Neither NIH download has a DOI, so the table cites the dataset paper and the NLM download page. The raw files are not copied into the repository: they sit in the project root, are

ignored by git, and are read only through the paths in `malaria/paths.py`. `NIH-NLM-ThinBloodSmearsPf` is the release that `cell_images` was cut from (Kassim et al., 2021). 192 of the 200 cell_images patients and 960 of the release's 965 photographs appear in both (`outputs/tables/nih_source_overlap.csv`, `scripts/check_nih_overlap.py`), so it is used only to recut NIH cells for the EDA and never as a test set.

Dataset	Source and persistent identifier	Role
NIH cell_images	Rajaraman et al. (2018), doi:10.7717/peerj.4568; download from the NLM malaria datasheet (downloaded 7 May 2026)	training, validation and hold-out
NIH-NLM-ThinBloodSmearsPf	Kassim et al. (2021), doi:10.1109/JBHI.2020.3034863; same NLM page (downloaded 15 Sept 2026)	EDA only (overlaps cell_images)
BBBC041 v1	Ljosa et al. (2012), doi:10.1038/nmeth.2083; Hung & Carpenter (2017), doi:10.1109/CVPRW.2017.112; bbbc.broadinstitute.org/BBBC041 (downloaded 7 May 2026)	external test (site_a, site_b)
MP-IDB	Loddo et al. (2019), doi:10.1007/978-3-030-13835-6_7; GitHub repository (downloaded 7 May 2026)	external test (four species)

2.2 Quantitative and qualitative description

The table gives the cell counts at the level the results are reported: by split, acquisition batch, species and life stage. NIH has 27,558 cells from 200 patients and 1,408 photographs. 150 patients contribute parasitised cells and 50 contribute only uninfected cells, and parasitised and uninfected cells are close to 50/50 in every split. BBBC041 is two batches that differ in image size and file format (1,208 images at 1200 × 1600 in PNG and 120 at 1383 × 1944 in JPG). The source does not say which batch came from which country, so they are labelled by observable batch, site_a and site_b. Only 2.7% (site_a) and 5.1% (site_b) of the cells are parasitised. MP-IDB annotates parasites only, so it has no uninfected cells and specificity cannot be measured on it. *P. falciparum*, the NIH species, accounts for 1,297 of its 1,437 parasites; the other three species together have 140.

Set	Detail	Images	Parasitised	Uninfected
NIH train	140 patients	958	9,619	9,584
NIH validation	30 patients	239	2,105	2,073
NIH hold-out	30 patients	211	2,055	2,122
BBBC041 site_a	ring 353, trophozoite 1,473, schizont 179, gametocyte 144	1,208	2,149	77,420
BBBC041 site_b	ring 169, trophozoite 111, schizont 11, gametocyte 12	120	303	5,614
MP-IDB *P. falciparum*	ring 1,230, trophozoite 42, schizont 18, gametocyte 7	104	1,297	n/a
MP-IDB other species	*P. malariae* 43, *P. ovale* 33, *P. vivax* 64	106	140	n/a

Table 1. Cells used in the primary metric, read from `data/manifests/` through `malaria.data.nih_cells` and `test_cells` (`malaria/data.py`). BBBC041 also has 446 `difficult` cells and 103 leukocytes, which are excluded (§3.1).

Qualitatively, the four sources look different before any model sees them (Figure 2). NIH cells are segmented and placed on a black background, pink and evenly lit. BBBC041 crops are rectangles cut from the smear: site_a cells are sharp, saturated and often blue-purple, site_b cells are grey and slightly blurred, and both show parts of neighbouring cells. MP-IDB crops are yellow to pink with a visible host cell, and most *P. falciparum* parasites are small rings.

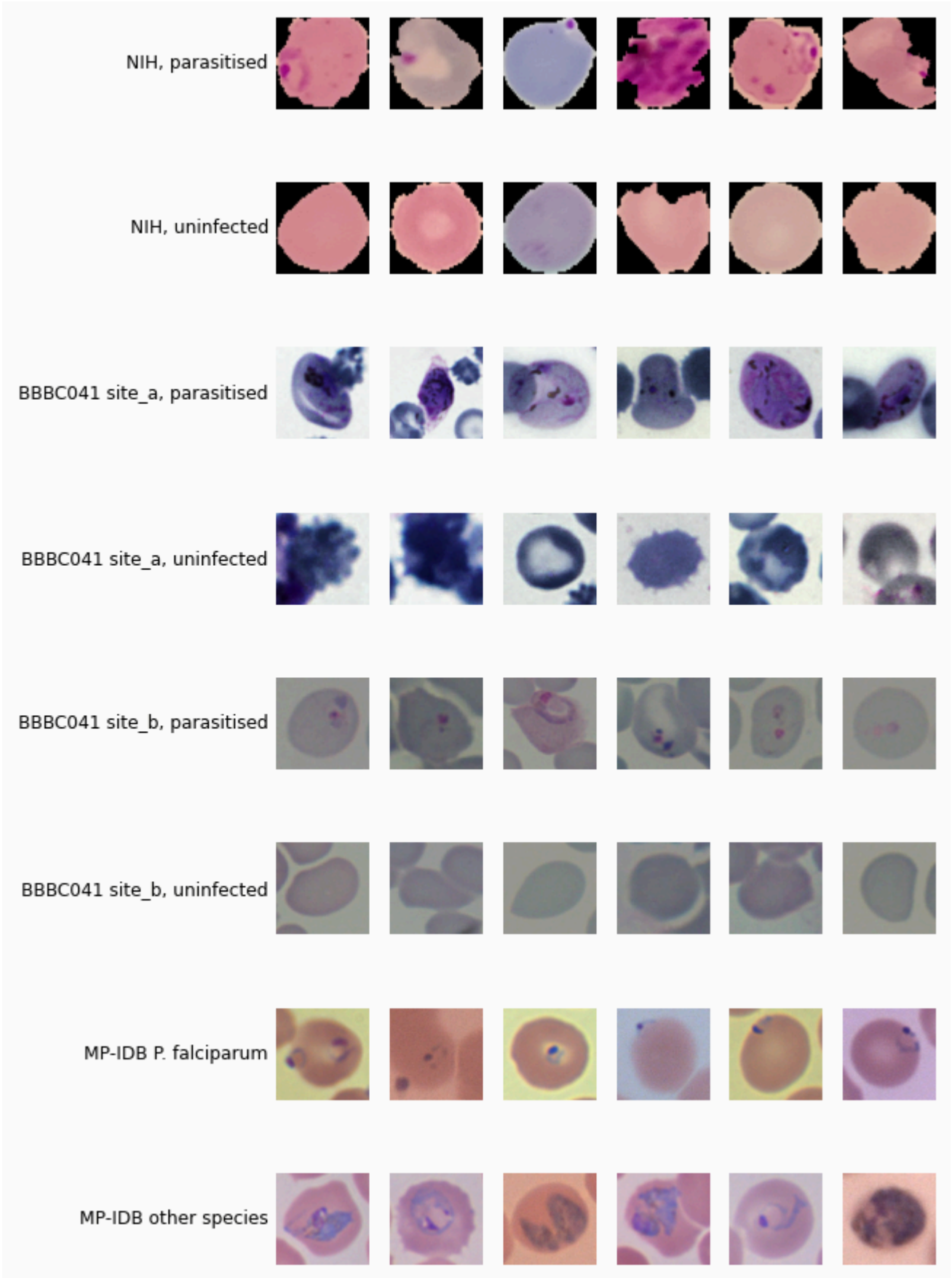


Figure 2. Random cells from every test set, as the models see them in RQ1. NIH cells sit on black; the external crops keep their background and parts of neighbouring cells.

2.3 Bias, representativeness and privacy (ELSA)

Privacy. The images are microscope photographs of blood cells. None of the three releases contains names, ages, sex or clinical data. NIH identifies patients only by a code (`C###`), which this project uses to group the split and for nothing else. No new data are collected.

Licences. BBBC041 is released under CC BY-NC-SA 3.0, which allows non-commercial use and requires derived data to be shared under the same terms. NIH-NLM-ThinBloodSmearsPf may be redistributed as long as NLM's notice is kept and NLM is credited, and the MP-IDB repository is MIT-licensed. The repository therefore holds code, manifests and summary tables only. No image or crop is redistributed, and the Drive copies used for Colab are private and will be deleted after the project.

Protected categories. The data carry no demographic variables, so results cannot be broken down by sex, age or ethnicity. This is a limitation, not a test that passed: a model could still work worse for some patient groups, and this project could not show it. The subgroups that can be analysed are the ones in Table 1, which are also the RQ3 subgroups: acquisition batch, species and life stage.

Representativeness. NIH comes from one hospital, one staining protocol and one smartphone-on-microscope setup, and it has a single species. Its 50/50 class balance is a result of curation: in the source photographs only 3.3% of annotated cells are parasitised, close to site_a's 2.7%. A model tuned to NIH's balance can therefore look better on the hold-out set than it would at clinical prevalence, which is why sensitivity and specificity are reported separately and accuracy is not. The external sets add a second species (*P. vivax*, BBBC041) and a European reference laboratory (MP-IDB), but no smears from sub-Saharan Africa, where most malaria deaths occur. The generalisation claims are limited to these three sources.

Mitigations (ELSA Part B). The D5 checklist (deon base items plus project items) is in the ELSA report, `docs/elsa_report.md`, with answers to the panel-review questions. The table repeats its Part B: the mitigation for each item judged critical.

Item	How the project deals with it
A.2 Collection bias	Each source is treated as a separate domain and reported apart: site_a and site_b, and *P. falciparum* against the other species. There is no single pooled score that one site could dominate. The claims are limited to these three sources.
A.4 No protected attributes	This is stated as a limitation, not as a test that passed: disparities by sex, age or ethnicity cannot be measured. The subgroups that can be measured (acquisition batch, species, life stage) are analysed instead in RQ3.
A.5 Licences	Only code, manifests and summary tables go into the repository. Raw images and crops stay out of git, and so do the Drive copies, which are kept for this non-commercial study only. BBBC041's share-alike term therefore never applies to a redistributed file, and each dataset paper is cited and NLM credited.
A.6 Label quality	BBBC041's annotators marked 446 cells as `difficult`; these and the 103 leukocytes stay in the manifests but are left out of the primary metric. For each of MP-IDB's 210 images, the tracked `mpidb_stage_audit.csv` records how many parasites it holds, how many crops were shipped and where the stage labels came from. NIH labels are used as released; checking them again is out of scope and is stated as a limitation.
A.7 Overlapping releases	192 of the 200 cell_images patients and 960 of the release's 965 photographs appear in both NIH releases (`nih_source_overlap.csv`), so NIH-NLM-ThinBloodSmearsPf is never used as a test set. It is used only to recut NIH cells for the EDA, and every recut cell inherits its patient's split (`inherit_nih_split`), so no patient crosses splits even there.

Item	How the project deals with it
C.2 Prevalence mismatch	The NIH classes are not rebalanced, because the model trains at NIH's 50/50 split. The mismatch is handled at evaluation instead: sensitivity and specificity are reported separately, and neither depends on prevalence. Accuracy is not reported for BBBC041, and F1 is reported only as a secondary measure.
C.6 Patient leakage	The NIH split is by patient, pinned in `nih_split.csv`, and checked before every training run. The overlapping NIH-NLM release is never used as a test set.
D.2 Disparate error rates	Sensitivity is compared across species, with an interval on the difference, and across BBBC041 life stages, as share of misses and miss rate. Specificity is compared across the two BBBC041 batches. Every interval comes from resampling source images.
D.3 Metric choice	A missed parasite (a patient left untreated) and a false alarm (unnecessary antimalarials) cost different things, so they are reported apart. The threshold is fixed at 0.5 before testing. AUC is only a threshold-free check.
D.4 Explainability	Saliency methods such as Grad-CAM are out of scope, as the proposal already stated. RQ2 is a controlled test of what the models depend on: if removing the background or matching colour changes their decisions, the models were using those cues.
D.8 Shortcut learning	RQ2's first step puts the test cells into the NIH black-frame format. The share of the drop this recovers shows how much the models relied on the frame instead of the parasite.
E.1 Regulation and redress	This is a research prototype, trained and tested on public data and never used on patients. The EU AI Act excludes AI systems developed only for scientific research (Art. 2(6)). A deployed diagnostic version would be a medical device (MDR 2017/745) and a high-risk AI system, requiring clinical validation, which this project does not do.
E.5 Untested populations	The three sources are a hospital in Chittagong (NIH), the two acquisition batches of BBBC041 and a laboratory in Lausanne (MP-IDB); none is from sub-Saharan Africa. Every claim is limited to these sources, and each is reported on its own so that no pooled score hides one of them. The gap is stated as a limitation in the Methodology Overview and the final report.
E.6 Automation bias	The models are research prototypes and are never used on patients; the README, the model information sheet and this report say so. Each output is a probability cut at a fixed 0.5 threshold, not a diagnosis, and the seed-0 results show how often it is wrong on new data: sensitivity on MP-IDB *P. falciparum* is between 28.6% and 57.4% across the three models. Any clinical use would need a microscopist to confirm every call and a separate clinical validation.

Table 2. ELSA Part B mitigations, copied from the ELSA report, where each row also says where it is implemented.

3. Dataset cleaning, preprocessing and splitting (stages B and C)

3.1 Cropping, manifests and exclusions (stage B)

Scripts: `scripts/build_crops.py`, `malaria/manifests.py`, `malaria/crops.py`, `malaria/background.py`

Every cell, from any dataset, becomes a row in a manifest with one shared schema (`manifests.COLUMNS`): dataset, split, cell ID, crop path, binary label, evaluation group, stage, species, patient, source image and crop box. Later stages read cells only through these tables. The steps are applied in this order:

Step	Operation	Justification
1	Build one manifest per dataset (`build_nih_manifest`, `build_bbbc041_manifest`, `build_mpidb_wholecell_manifest`)	One schema means every later stage treats the datasets the same way.

Step	Operation	Justification
2	Cut BBBC041 cells with one rule (`crops.square_padded_box`): a square around the annotated box, padded by 10%, clipped and flagged where it meets the image border	A square crop matches the square NIH input; the padding keeps the whole cell edge in view.
3	Recentre MP-IDB crops on the red cell that holds each parasite (`background.host_cell_box`, `build_crops.py --mpidb-wholecell`). The host cell was found for 1,091 of the 1,437 parasites; the other 346 (340 *P. falciparum*: 338 in cell clumps and 2 missed; 6 *P. malariae*) fall back to a cell-sized square centred on the parasite (`mpidb_wholecell_framing.csv`)	MP-IDB annotates parasites, not cells, and NIH crops show a whole cell. The shipped close-ups cut off the cell edge and mix masked and unmasked formats by species.
4	Exclude BBBC041 `difficult` cells (446) and leukocytes (103). They stay in the manifest with `eval_group` set to `excluded_*`, and `data.test_cells` returns only `primary` rows	`difficult` cells are ambiguous by the annotators' own label, and leukocytes are a class the model never saw (the NIH negative class is uninfected red cells).
5	Never use NIH-NLM-ThinBloodSmearsPf or the MP-IDB parasite close-ups (`mpidb`) as evaluation sets	The first overlaps cell_images (§2.1); every RQ evaluates whole cells.
6	Check for missing values	None among the primary cells: each has a path, a label and a source image, and `CellDataset` raises an error on an unlabelled row.

Output: `data/manifests/*_cells.csv` (rebuilt deterministically, so gitignored) and the crops under `data/crops/`, saved as PNG by dataset, split and class so they can be browsed in a file explorer (see the table in `README.md`). The reframing keeps every MP-IDB parasite, stage label and cell ID, which the build asserts.

3.2 Patient-grouped split (stage C)

Script: `malaria/splits.py`; pinned output `data/manifests/nih_split.csv`

NIH is split 70/15/15 by patient (`splits.patient_grouped_split`, seed 42). Patients are assigned greedily, largest first, to whichever split is furthest below its target number of cells, so no patient is split. The assignment does not target class balance, but every split stays close to 50/50 (Table 3). The split was computed once and written to the tracked `nih_split.csv`, so all three architectures train on the same rows. The external test sets are kept whole and never split: the models do not see them until stage G.

	Train	Validation	Hold-out
Cells	19,203	4,178	4,177
Patients	140	30	30
of which only uninfected cells	36	7	7
Photographs	958	239	211
% parasitised	50.1	50.4	49.2

Table 3. The three NIH splits are similar in size per patient, share of uninfected-only patients and class balance.

3.3 Data leakage

Kapoor and Narayanan (2023) define leakage as a spurious relationship between inputs and target that comes from the data collection, sampling or preprocessing. The table lists each route that applies here and how it is closed. The model information sheet in Appendix B summarises the training population and the external evaluation.

Leakage route	Risk here	Prevention
No clean train/test separation	cells of one patient share a slide, stain and lighting	split by patient; `check_no_patient_leakage` runs before every training job
Duplicates across splits	the same cell saved twice	exact hashes: none; perceptual-hash matches join different patients (EDA section B)
Test data in preprocessing	colour reference or threshold fitted on test cells	colour reference from NIH train only; threshold fixed at 0.5 before testing
Model selection on test data	choosing epochs, methods or thresholds by test score	early stopping on NIH validation only; both colour methods reported
Overlapping releases	NIH-NLM-ThinBloodSmearsPf photographs the same patients	never used as a test set; recut cells inherit their patient's split (`inherit_nih_split`)

3.4 Data-quality issues and continuous-analysis checks

Table 4 lists every oddity found while preparing the data and what was done about it.

Issue	Finding	Status
NIH cells on black padding	The median NIH crop is 26% black padding; no external crop has any.	Kept as the training format. The EDA measures tissue pixels only (§4), and RQ2 step F1 gives test cells the same frame (§5.2).
MP-IDB has no uninfected cells	Only parasites are annotated.	Specificity, AUC, precision and F1 are measured on NIH and BBBC041 only.
MP-IDB close-ups	The shipped crops show the parasite, cut off the cell edge, and are masked for some species and not for others.	Fixed: every parasite is reframed on its host red cell (§3.1, step 3).
Host cell not found	346 MP-IDB parasites (340 *P. falciparum*, 6 *P. malariae*) get a cell-sized square instead.	Kept; RQ3 is also reported without them (`rq3_<model>_s0_without_fallback.csv`).
Mask centre on background	For 1,029 of the 86,035 BBBC041 crops (1,022 of them primary cells), Otsu found no cell under the crop centre and kept the nearest region, which may be a neighbour.	Kept; RQ2 is also reported without these cells (`rq2_<model>_s0_without_nearest.csv`).
Ambiguous and out-of-scope cells	BBBC041 has 446 `difficult` cells and 103 leukocytes.	Excluded from the primary metric, kept in the manifests (§3.1, step 4).
Overlapping NIH releases	192 of 200 patients and 960 of 965 photographs are shared.	The NIH-NLM release is never a test set (§3.3).
Errors found in code review	A code review found that Reinhard transfer used a spread pooled over many cells, that histogram matching broke ties toward the dark end, and that the NIH bootstrap resampled photographs instead of patients.	Fixed before the results in §5.6; VGG-16 was re-scored on the corrected crops.
Missing values	None among the primary cells.	`CellDataset` raises an error on an unlabelled row.

Table 4. Data-quality issues found during preparation and their status.

There is no separate test suite. The checks run inside the pipeline (Beaulieu-Jones & Greene, 2017), so a run that breaks an assumption stops with an error instead of producing numbers. Among them: the NIH split sizes must equal the pinned 19,203 / 4,178 / 4,177; no patient may appear in two splits; every model parameter must be trainable; the reframing must keep every MP-IDB parasite; a perfect predictor must score 1 on every metric in `python -m malaria.metrics`; and models are compared only if they were scored on the same manifests, crop contents and cells. The full list is in Appendix C.

The build scripts must run in a fixed order, because each reads the output of the previous one: `build_crops.py` (manifests and crops), then `compute_stats.py` (EDA statistics), then `build_masked_crops.py` and `build_colour_crops.py` (RQ2 inputs), and only then training and evaluation (§6).

4. Exploratory data analysis (stage D)

The EDA (`notebooks/03_eda_short.ipynb`, full record in `01_eda.ipynb`) measured every cell on tissue pixels only, because a quarter of each NIH crop is black padding that would otherwise dilute brightness and colour averages (`malaria/imagestats.py`, `scripts/compute_stats.py`). Differences are reported as a rank-based separation between 0 (identical distributions) and 1 (no overlap), computed in `imagestats.separation`. Three findings decided the method.

Finding	Evidence	Consequence
Crop format is the largest difference (Figure 3)	The median NIH crop is 26% black padding, and no external crop has any. Separation on the fraction of black pixels is 1.00 against every test set.	RQ2 step F1 removes the background of the test crops.
Stain colour differs, in a different direction per test set (Figure 4)	After Giemsa colour deconvolution, BBBC041 cells absorb three to four times as much methyl blue as NIH cells (median 0.065 to 0.090 against 0.022). On the red-minus-blue axis NIH sits close to MP-IDB *P. falciparum*, while BBBC041 sits at or below zero. A hue and saturation shift toward NIH moves methyl blue further away (separation 0.88 against 0.72); Reinhard transfer or histogram matching brings both stain channels close to NIH (`colour_transform_effect.csv`).	RQ2 step F2 uses Reinhard transfer and histogram matching, not an HSV shift.
Sharpness differs in both directions	site_a and *P. malariae* are sharper than NIH; site_b, *P. ovale* and *P. vivax* are blurrier.	A blur sweep cannot reproduce the shift, so the proposal's blur, noise and contrast experiments were dropped from RQ2.

Table 5. The three EDA findings that shaped the method.



Figure 3. Crop format. NIH cells are segmented and placed on black; the external crops are rectangles cut from the smear.

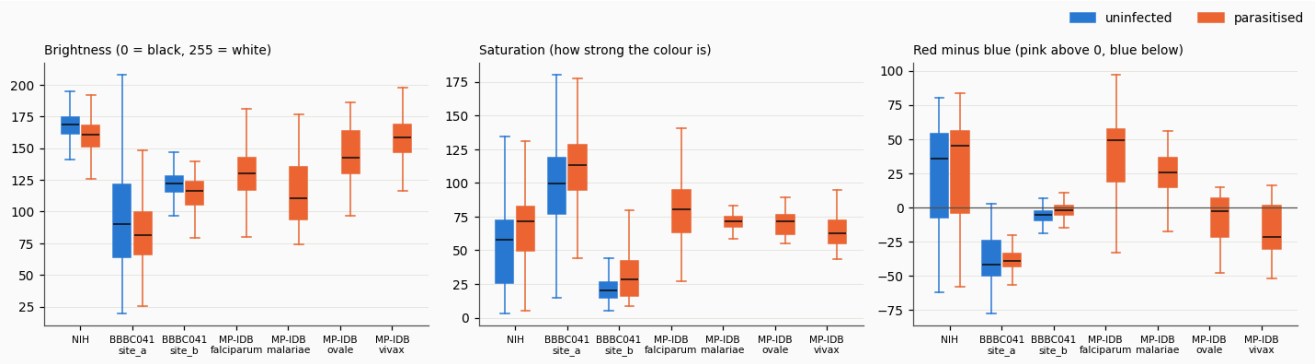


Figure 4. Tissue brightness, saturation and red minus blue per test set and class (boxes: quartiles; whiskers: 5th to 95th percentile).

Individual cells were inspected as well as distributions: Figure 2 shows random cells per set and class, and the full EDA record adds contact sheets and the smallest crops of each set, which were checked by eye for cut-off cells.

5. Models and evaluation (stages E to H)

5.1 Model arms and baseline protocol (stage E)

Scripts: `malaria/models.py`, `malaria/data.py`, `scripts/train.py`, `notebooks/10_train_colab.ipynb`

RQ1 compares one lightweight and two heavy architectures, all built by `models.build_model` from torchvision's ImageNet weights (Russakovsky et al., 2015), with the 1,000-class head replaced by a single output. They are the three networks named in the proposal, and they span a sixty-fold range in size while sharing an input size and a pretraining dataset.

Arm	Model	Parameters (with new head)	Rationale
E1	VGG-16 (Simonyan & Zisserman, 2015)	134.3 M	Plain stacked convolutions; the oldest and largest network
E2	ResNet-50 (He et al., 2016)	23.5 M	Residual network; the standard heavy contender

Arm	Model	Parameters (with new head)	Rationale
E3	MobileNetV2 (Sandler et al., 2018)	2.2 M	The lighter architecture RQ1 asks about

Following the literature review (§2.3), the baseline is the patient-grouped protocol of Hou et al. (2026) applied to all three architectures, not one model.

- *Architecture.* torchvision weights `IMAGENET1K_V1` for all three, so the pretraining recipe is the same generation. The last layer becomes a single-output linear layer whose output is the logit for *parasitised*: `classifier[6]` in VGG-16, `fc` in ResNet-50, `classifier[1]` in MobileNetV2.
- *Fine-tuning.* Every parameter is trained (VGG-16 134,264,641; ResNet-50 23,510,081; MobileNetV2 2,225,153); no layer is frozen, and `check_all_trainable` asserts this before and after training.
- *Input.* Each crop is padded to a square with black, resized to 224 × 224 and normalised with the ImageNet channel means and standard deviations (`data.pad_to_square`, `data.load_input`).
- *Training.* Binary cross-entropy on the logit, Adam (Kingma & Ba, 2015), mixed precision on the GPU.
- *Output.* `models/<model>_s<seed>.pt`, a per-epoch log (`outputs/tables/<model>_s0_training_log.json`) and a resumable checkpoint written after every epoch. A resumed run must have the settings it started with, and keeps the commit it started at.
- *Where it runs.* A Colab T4 GPU through `notebooks/10_train_colab.ipynb`, which unpacks the archive built by `scripts/pack_for_colab.py`. On the laptop, `train.py --smoke` checks the same code on 64 cells.

Class balance and resampling. NIH training cells are 50.1% parasitised, so the training data are neither undersampled nor oversampled, and the loss is not reweighted. The imbalance that matters is at test time (2.7 to 5.1% on BBBC041). It is handled by reporting sensitivity and specificity separately, since neither depends on prevalence.

5.2 RQ2 test inputs (stage F)

Scripts: `scripts/build_masked_crops.py`, `scripts/build_colour_crops.py`, `malaria/colour.py`

RQ2 changes only the test crops; the trained models stay fixed. The two steps follow from the EDA (§4).

Step	Operation	Justification
F0	Raw crops from §3.1, background kept	The RQ1 input.
F1	Threshold each test crop on brightness with Otsu's method (Otsu, 1979). Keep the connected region under the crop centre, or the nearest one when the centre falls on background (1,029 of 86,035 BBBC041 crops); for MP-IDB keep the region that holds the annotated parasite and add the parasite pixels back. Set everything else to black and cut the result to the square around the kept cell. All 87,472 test crops were processed and no mask came back empty	This is how cell_images crops are cut. Otsu reached a median Dice of 0.90 against the NIH hand-drawn outlines, while rembg U²-Net found no cell in 21% of NIH crops (`background_method_summary.csv`). Adding the parasite back means no parasite is removed (median retention before that step 0.98 to 1.00).
F2a	Reinhard transfer (Reinhard et al., 2001): match the mean and standard deviation of each Lab channel to the average per-cell mean and standard deviation of NIH cells (`colour.reinhard`)	It rescales each cell's own spread; a spread pooled over many cells would add the differences *between* NIH cells and inflate every test cell's contrast.

Step	Operation	Justification
F2b	Histogram matching: map each RGB channel's quantiles to the pooled NIH quantiles (`colour.match_to_reference`)	A second, non-parametric way to reach the NIH colour distribution.

Both F2 methods use tissue pixels only and start from the F1 crops. The NIH reference comes from 1,000 randomly drawn *training* cells (seed 0), saved to `outputs/tables/colour_reference.json`, so no hold-out cell affects the test inputs. Both methods are reported; neither is chosen by its test score.

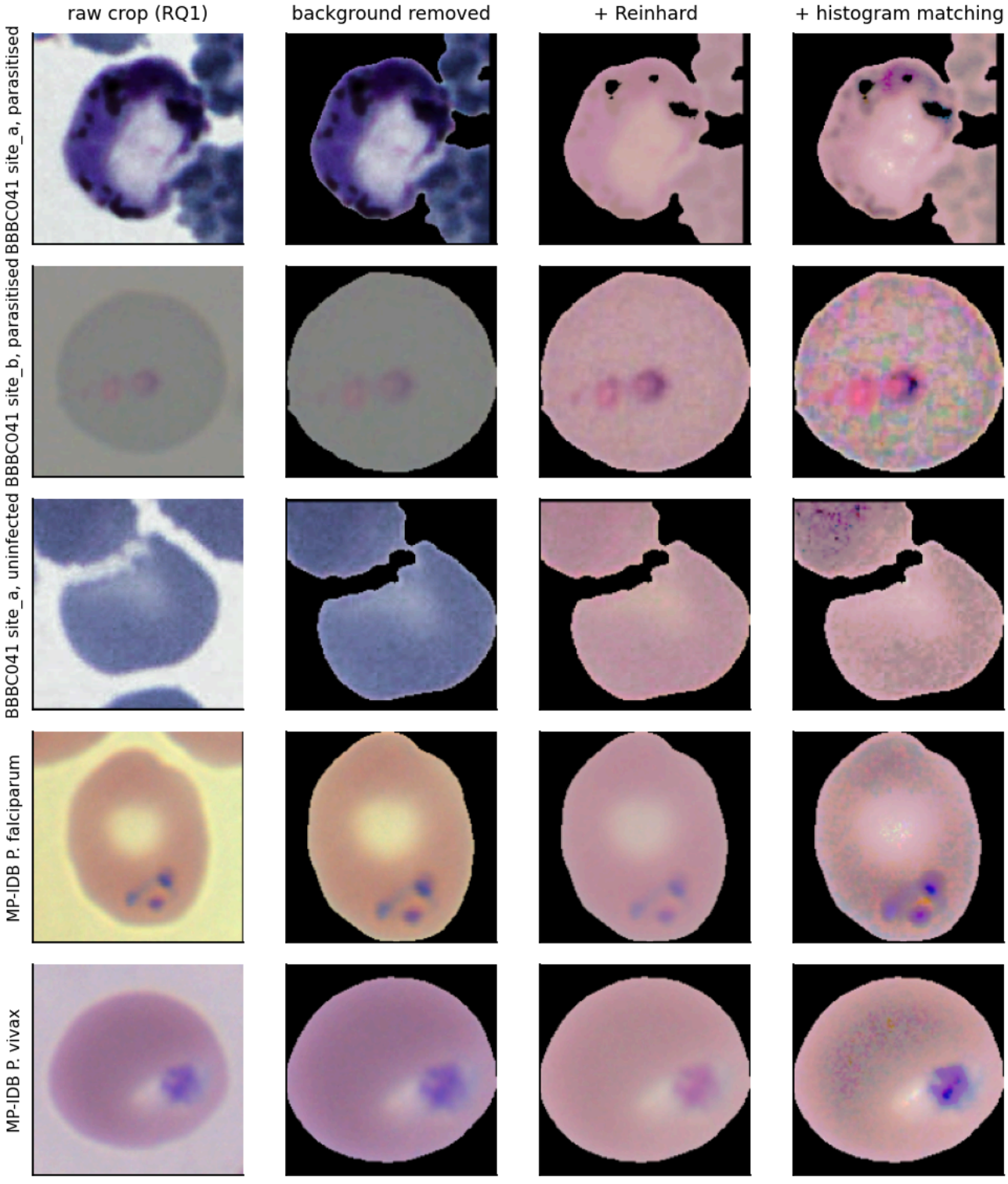


Figure 5. The same test cells as RQ1 and RQ2 see them. Three known artefacts are visible. On BBBC041, Otsu keeps a touching neighbour together with the cell (rows 1 and 3). Very dark site_a pixels

fall below the black level and are treated as padding (row 1). Histogram matching stretches the small colour spread of the grey site_b cells and amplifies their noise (row 2).

5.3 Hyperparameters

All values are fixed in advance (`SETTINGS` in `train.py`, `data.py`, `models.py`) and are the same for the three arms; `--resume` refuses a run whose settings changed. *Importance* says how much changing the value would be expected to change the RQ results.

Hyperparameter	Value	Importance	Why this value
Learning rate	1×10^{-4}	high	Sets how far the pretrained weights move. Loddo et al. (2022) fine-tuned the same three networks on NIH at this rate and found smaller rates learned too slowly and larger ones did not converge; Rajaraman et al. (2019) trained all layers at the same rate.
Epochs and early stopping	at most 20; stop when NIH validation loss has not improved for 3 epochs; keep the best-validation weights	high	Decides how closely each model fits the NIH training patients. It replaces the fixed ten epochs of Loddo et al. (2022) and guards against training too long. All three models stopped early (Figure 6).
Decision threshold	0.5 (`metrics.THRESHOLD`), fixed before any test set was seen	high	Sets the balance between sensitivity and specificity, the reported measures. At seed 0 the architectures differ mostly in where their scores fall relative to it (§5.6).
Pretrained weights and trained layers	torchvision `IMAGENET1K_V1`; every layer trained	high	Stain and cell texture are far from ImageNet photographs, so the early layers need to adapt too; a frozen network with a new head (Rajaraman et al., 2018) can only reweight ImageNet features.
Batch size	32	medium	Affects gradient noise and memory. 32 follows Loddo et al. (2022) and fits VGG-16 on a T4.
Augmentation	horizontal and vertical flips, rotation by multiples of 90° (`data.augment`); no colour jitter	medium	A cell turned upside down is still the same cell, so these change neither colour nor the black frame. Colour jitter is left out on purpose: a model trained to ignore colour would leave less for RQ2's colour step to recover, by an amount set by an arbitrary jitter strength. Hou et al. (2026) trained with colour jitter and still saw specificity fall to 18.0% on BBBC041.
Input	224 × 224, black padding to square, ImageNet mean and standard deviation	medium	Matches the pretrained weights; black padding matches how NIH crops are stored.
Optimiser	Adam, betas (0.9, 0.999), weight decay 0	low	PyTorch defaults, as in Loddo et al. (2022); not varied.
Loss	`BCEWithLogitsLoss`, no class weights	low	The training data are 50.1% parasitised, so there is nothing to reweight.
Dropout	torchvision defaults: VGG-16 0.5, MobileNetV2 0.2, ResNet-50 none	low	Each network is used as published.
Mixed precision	on the GPU only	low	Speeds up training; the model and loss are unchanged.

Hyperparameter	Value	Importance	Why this value
Seeds	training 0 (0 to 2 planned); split 42; colour reference 0; bootstrap 0	low per value	One seed fixes one run; the final results report the spread over three training seeds per architecture.

Table 6. Every hyperparameter, its value and its expected importance.

No grid search or Optuna is used. Tuning each architecture separately would make the RQ1 comparison partly a comparison of tuning budgets, and a free Colab GPU does not allow a search over three large networks. Only the seed varies, so that each final result is a distribution over runs instead of one run.

5.4 Fair model comparison

The following are held constant across the three arms and across the RQ2 variants:

- the NIH train, validation and hold-out rows in `nih_split.csv`;
- the input pipeline (black padding to square, 224 px, ImageNet normalisation) and augmentation;
- the optimiser, learning rate, batch size and stopping rule;
- the test cells: the same primary rows of NIH hold-out, BBBC041 site_a and site_b, MP-IDB *P. falciparum* and other species;
- the decision threshold of 0.5, fixed before any test set was seen;
- the metric code and the bootstrap: each set's resamples are keyed by its name, so every model and variant is resampled on the same NIH patients and test photographs;
- the code commit and the inputs, checked: `run.json` records, per prediction file, the checkpoint, the manifest hashes and a digest of the crops read, and `compare_models.py` refuses models whose inputs differ.

What differs between arms is the architecture, and what differs between RQ2 variants is the test input. This is what lets a difference in drop be attributed to one of the two.

5.5 Evaluation strategy (stages G and H)

Scripts: `scripts/evaluate.py`, `malaria/metrics.py`, `scripts/summarise_results.py`, `scripts/compare_models.py`

Metrics. All definitions follow Table 2 of the literature review. The task is binary classification with parasitised as the positive class. *Sensitivity* is the share of parasitised cells called parasitised, and *specificity* the share of uninfected cells called uninfected. Specificity is measured on NIH and BBBC041 only, since MP-IDB has no uninfected cells. AUC is a threshold-free check on NIH and BBBC041. The *drop* is the NIH hold-out value minus the test-set value, in percentage points, per model and test set (`metrics.drop_pp`).

Why not accuracy or F1 as the main metric. At 2.7% prevalence, a model that calls every BBBC041 cell uninfected scores 97.3% accuracy. F1 combines sensitivity with precision, and precision falls with prevalence even when the model does not change: Hou et al. (2026) show that a single operating point gives a precision of 15 to 27% below 1% prevalence and 66 to 90% at 5 to 20%. F1 is therefore reported as a secondary measure, with precision next to it and the prevalence of each set, on NIH and BBBC041 only (`metrics.precision_f1`); no drop in F1 is computed, because the NIH hold-out set is 49.2% parasitised and site_a 2.7%. On MP-IDB precision is 100% by construction and F1 would only repeat sensitivity. A high AUC can also sit alongside an unusable operating point: Zhao et al. (2020) report an AUC of 0.945 on BBBC041, while Hou et al. (2026) found 18.0% specificity at their threshold.

Uncertainty. Every value has a 95% percentile interval from a bootstrap that resamples *source images*, not cells (`metrics.bootstrap`, 2,000 resamples for every metric). Cells cut from one photograph share its stain, focus and lighting, so treating them as independent would make the intervals too narrow. The NIH hold-out set is resampled one level higher, by *patient* (30 patients, `metrics.reference_draws`), since a patient's photographs come from one slide. The test sets have no patient identifiers, so they are resampled by photograph. The interval for a drop comes from resampling the NIH hold-out set and the test set independently and taking the difference of the draws. Each set's random stream is keyed by its name (`metrics.resample_weights`): different sets are resampled independently, while the same set is resampled identically for every model and variant, which makes differences between models paired.

Model comparison (RQ1). All arms are scored by `evaluate.py` on the same test sets, with the same cells and the same threshold. For every pair of models, test set and metric, `compare_models.py` gives the difference in drop with a paired bootstrap interval. With three models that is 24 comparisons. They share one 5% error budget (Bonferroni correction), so each interval is widened from 95% to 99.8%, and a difference counts as clear only when this adjusted interval excludes zero. Without the correction, about one of 24 comparisons could look clear by chance alone.

Ablation (RQ2). The frozen models score each test set on F0, F1 and both F2 variants. The *share recovered* is (value after a step − value before it) divided by the RQ1 drop, reported for each step and in total (`metrics.share_recovered`). It is left undefined unless the raw drop is clearly above zero, i.e. the lower end of its 95% interval is positive: with no measurable drop there is nothing to recover, and dividing by a drop near zero gives meaningless shares, such as −62 on a 1.5-point drop whose interval includes zero.

Error analysis (RQ3). `summarise_results.py` breaks down the misses of each model. Sensitivity on pooled other-species MP-IDB cells is compared with *P. falciparum*, with an image-resampled interval on the difference. For BBBC041, the table gives each life stage's share of all missed parasites (the RQ3 wording) together with its miss rate. The miss rate is shown because the shares follow the stage counts: trophozoites are 69% of site_a parasites, so they would account for most misses even at an average miss rate.

Out-of-sample testing. The NIH hold-out patients are used once, after training, and BBBC041 and MP-IDB are never used for training, early stopping, the colour reference or the threshold. Both external sets are therefore out-of-distribution tests in the sense the MRQ needs.

5.6 Baseline results (seed 0)

The three arms were trained on a Colab T4 (VGG-16 on 30 September, ResNet-50 and MobileNetV2 on 2 October 2026) and scored on the same crops: one code commit and identical crop digests in each `outputs/tables/<model>_s0_run.json`. These are single-seed results: they show that the pipeline works and how large the effects are, not the final answer to any RQ.

Convergence

Model	Epochs run	Best epoch	NIH validation loss	Validation sensitivity / specificity	Minutes per epoch (T4)
VGG-16	11	8	0.078	98.2% / 96.2%	3.2
ResNet-50	9	6	0.076	96.6% / 97.6%	1.8
MobileNetV2	9	6	0.077	96.5% / 98.0%	1.3

Table 7. Training runs, from `outputs/tables/<model>_s0_training_log.json`.

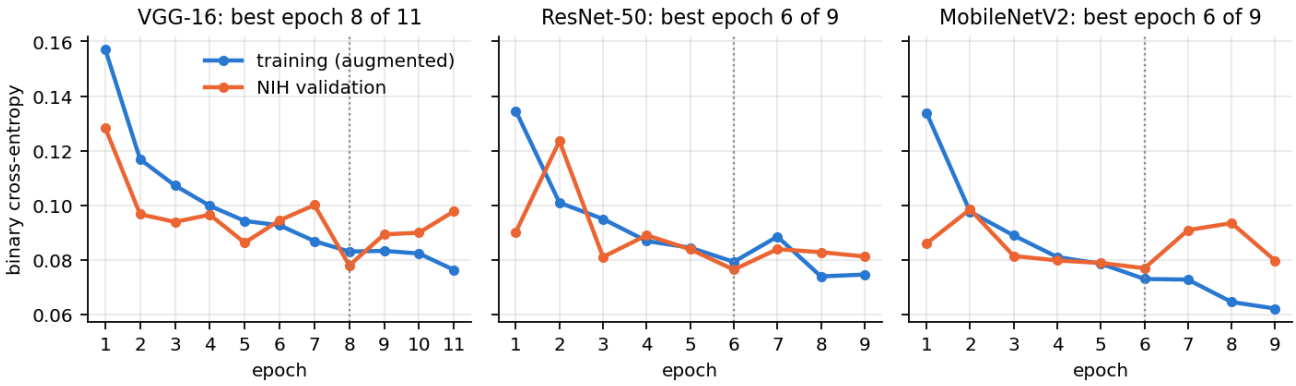


Figure 6. Training and NIH validation loss per epoch; the dotted line marks the best epoch, whose weights are kept. Training loss is measured on augmented cells during the epoch.

All three stopped early, and their best NIH validation losses differ by less than 0.002, so the three arms enter the external tests equally well fitted to NIH. After the best epoch, validation loss rose for all three while training loss mostly kept falling, the usual sign that further epochs would only fit the training patients more closely.

RQ1: how large is the drop, and does the lighter architecture lose less?

Test set, metric	VGG-16	ResNet-50	MobileNetV2
NIH hold-out sensitivity	98.2 (96.8 to 99.3)	97.2 (95.6 to 98.5)	97.1 (95.3 to 98.2)
NIH hold-out specificity	96.6 (94.7 to 97.9)	98.4 (97.8 to 98.9)	98.2 (97.5 to 98.8)
site_a sensitivity	99.3 (drop −1.1)	79.3 (drop 17.9)	83.7 (drop 13.4)
site_a specificity	65.5 (drop 31.0)	87.3 (drop 11.0)	90.0 (drop 8.1)
site_a AUC	97.2 (drop 2.5)	92.0 (drop 7.6)	94.5 (drop 5.1)
site_b sensitivity	89.8 (drop 8.5)	68.0 (drop 29.2)	30.4 (drop 66.7)
site_b specificity	95.0 (drop 1.5)	98.1 (drop 0.2)	99.1 (drop −0.9)
site_b AUC	97.5 (drop 2.2)	96.4 (drop 3.2)	93.0 (drop 6.6)
MP-IDB *P. falciparum* sensitivity	57.4 (drop 40.9)	52.6 (drop 44.6)	28.6 (drop 68.5)
MP-IDB other species sensitivity	80.0 (drop 18.2)	56.4 (drop 40.8)	38.6 (drop 58.5)

Table 8. Values in %, fixed threshold 0.5; drops in percentage points from the same model's NIH hold-out value. Intervals for every value and drop are in `rq1_vgg16_s0.csv`, `rq1_resnet50_s0.csv` and `rq1_mobilenet_v2_s0.csv`.

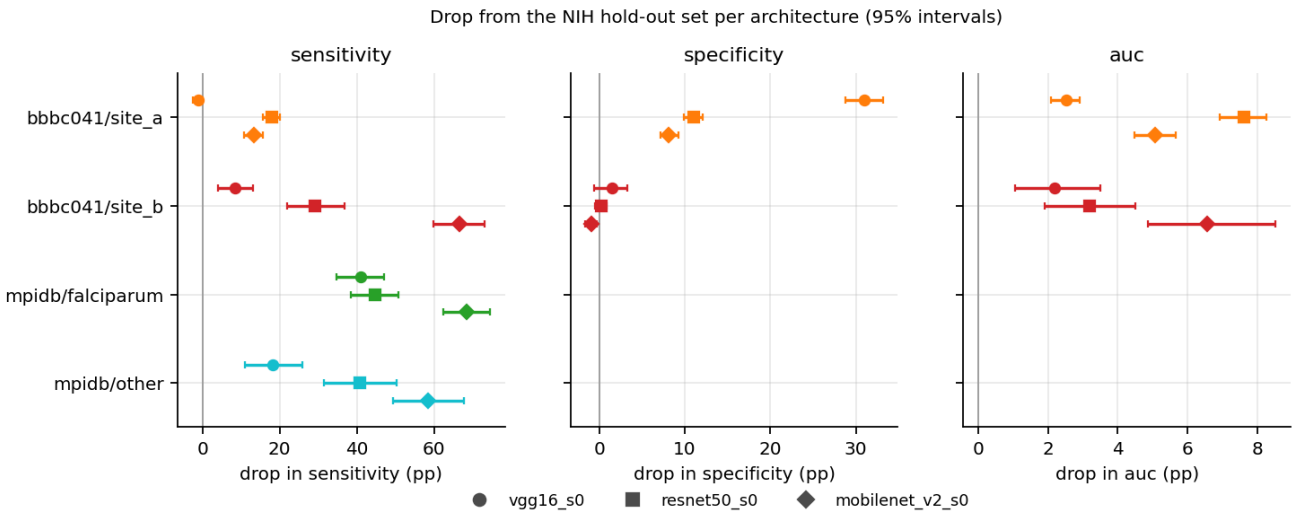


Figure 7. Drop from the NIH hold-out set per architecture, with 95% intervals (`compare_models.py`).

All three models do equally well on NIH (AUC 0.996 to 0.997), and all three lose on at least one metric on every external set, but they lose in different places. MobileNetV2, the lighter architecture, does not lose less. It loses the most sensitivity on site_b and on both MP-IDB groups, by 17.7 to 58.2 points more than the heavier models, and every one of these differences stays clear of zero after the Bonferroni adjustment (for example 27.6 points more than VGG-16 on *P. falciparum*, adjusted interval 20.5 to 34.4). On site_a the order reverses: VGG-16 loses 31.0 points of specificity, 22.9 more than MobileNetV2 (adjusted 19.8 to 25.9). VGG-16 has the smallest AUC drop on both BBBC041 sites. Of the 24 pairwise differences, 20 stay clear of zero after adjustment (`rq1_compare_s0.csv`).

Much of the difference between the architectures lies in where their scores fall relative to the 0.5 threshold. VGG-16 calls 34% of BBBC041 cells parasitised, against 14% for ResNet-50 and 11% for MobileNetV2, which is why it keeps its sensitivity and loses specificity while the other two do the opposite. AUC, which ignores the threshold, still favours VGG-16 on both sites. The answer to RQ1's second part is therefore no at the fixed threshold, with the caveat that part of the lighter model's loss is a shift of its scores toward *uninfected* on the external sets.

Secondary measure: precision and F1

Set	Measure	VGG-16	ResNet-50	MobileNetV2
NIH hold-out (49.2% parasitised)	precision	96.5 (90.6 to 98.4)	98.3 (95.9 to 99.1)	98.1 (95.2 to 99.0)
	F1	97.4 (94.1 to 98.5)	97.7 (96.0 to 98.6)	97.6 (95.5 to 98.4)
site_a (2.7% parasitised)	precision	7.4 (7.0 to 7.8)	14.8 (13.6 to 16.2)	18.9 (17.2 to 20.7)
	F1	13.8 (13.0 to 14.5)	24.9 (23.1 to 26.9)	30.8 (28.5 to 33.3)
site_b (5.1% parasitised)	precision	49.5 (44.2 to 55.4)	66.2 (60.9 to 71.8)	63.9 (55.7 to 72.3)
	F1	63.8 (59.0 to 68.7)	67.1 (61.6 to 71.9)	41.2 (33.9 to 47.7)

Table 9. Precision and F1 in %, with 95% intervals from resampling source images (NIH hold-out: patients), from `rq1_f1_<model>_s0.csv`. Not computed on MP-IDB, which has no uninfected cells.

F1 is between 97.4 and 97.7% on the NIH hold-out set for all three models and falls on BBBC041, most on site_a, where precision is 7.4 to 18.9%. At 2.7% prevalence, VGG-16's site_a specificity of 65.5% still means 26,684 false alarms against 2,134 parasites found, 12.5 for every parasite. F1 puts VGG-16 last

on site_a although its site_a AUC is the highest of the three: at this prevalence its false alarms outnumber the parasites, so its F1 is set mostly by how many cells it calls parasitised at the fixed threshold. This is why F1 is reported next to sensitivity and specificity, not instead of them.

RQ3: species and life stages

Model	*P. falciparum* minus other species (pp, 95% CI)	site_a stage with most misses (share; miss rate)	site_b stage with most misses (share; miss rate)
VGG-16	−22.6 (−31.7 to −13.1)	ring (93%; 4.0%)	ring (71%; 13.0%)
ResNet-50	−3.8 (−14.9 to 7.7)	trophozoite (55%; 16.5%)	ring (64%; 36.7%)
MobileNetV2	−10.0 (−21.0 to 0.7)	trophozoite (46%; 11.0%)	ring (52%; 65.1%)

Table 10. From `rq3_<model>_s0.csv`.

Only VGG-16 shows a species gap larger than its interval, and it runs the opposite way to a species effect: *P. falciparum*, the NIH species, is recognised less often. 1,230 of the 1,297 MP-IDB *P. falciparum* parasites are rings. On BBBC041 site_b, rings account for the largest share of misses for all three models (52 to 71%). On site_a, trophozoites (69% of the parasites) account for the largest share of misses for ResNet-50 and MobileNetV2, while their miss rate stays below that of rings and gametocytes. Leaving out the 346 fallback crops lowers *P. falciparum* sensitivity for all three models (to 53.2, 46.3 and 23.4%) and widens the gap to the other species (−27.4, −10.4 and −16.9 points). MobileNetV2's gap then also lies outside its interval (−28.3 to −5.2), so for MobileNetV2 the species result depends on these crops (`rq3_<model>_s0_without_fallback.csv`).

Rings are missed more often than trophozoites in five of the six model and site combinations; the exception is MobileNetV2 on site_b, which misses 65% of rings and 80% of trophozoites. This fits their size: a ring is a chromatin dot of 1 to 2 µm, a few pixels in a resized cell crop, against 5 to 8 µm for a mature trophozoite (Hou et al., 2026). Since most MP-IDB *P. falciparum* parasites are rings, a likely reading of VGG-16's species gap is the stage mix rather than the species. The final analysis tests this directly by comparing ring sensitivity across the two MP-IDB groups, for each model and seed.

RQ2: background removal and colour matching

Test set, metric	Model	Raw	Background removed	+ Reinhard	+ Histogram matching
MP-IDB *P. falciparum* sensitivity	VGG-16	57.4	71.9	66.4	99.5
	ResNet-50	52.6	63.9	52.3	98.1
	MobileNetV2	28.6	27.2	32.9	95.4
site_b sensitivity	VGG-16	89.8	90.8	89.8	100.0
	ResNet-50	68.0	73.6	75.9	95.0
	MobileNetV2	30.4	36.6	47.2	80.5
site_a specificity	VGG-16	65.5	53.6	74.9	5.9
	ResNet-50	87.3	66.6	82.0	6.6
	MobileNetV2	90.0	78.9	96.3	44.4
site_a AUC	VGG-16	97.2	96.6	85.6	91.6
	ResNet-50	92.0	90.2	74.5	72.7

Test set, metric	Model	Raw	Background removed	+ Reinhard	+ Histogram matching
	MobileNetV2	94.5	89.6	84.1	88.5

Table 11. Values in %, for the four rows that carry the RQ2 findings; every external set and metric is in Appendix A, and the shares recovered and intervals in `rq2_<model>_s0.csv`.

Removing the background raises MP-IDB *P. falciparum* sensitivity for VGG-16 (recovering 36% of its drop) and ResNet-50 (25%), but not for MobileNetV2. It also raises site_b sensitivity for all three models (by 1.0 to 6.2 points), while it lowers site_a specificity for all three. Reinhard transfer raises site_a specificity for VGG-16 and MobileNetV2 but costs site_a sensitivity, down to 15.9% for MobileNetV2. Histogram matching pushes almost every cell to *parasitised* for all three models: sensitivity is 80 to 100% on every set, while specificity falls to 5.9 to 44.4% on site_a and 1.1 to 41.9% on site_b. Compared with the raw crops, no step raises AUC on either BBBC041 site for any model. Leaving out the 1,022 BBBC041 cells whose mask fell back to a neighbouring region changes no BBBC041 value by more than 2.2 points for any model (`rq2_<model>_s0_without_nearest.csv`).

At the fixed threshold, the colour steps mostly move the models' scores instead of improving how well they separate the classes, and this holds for all three architectures. On MP-IDB, which has no uninfected cells, a higher sensitivity cannot tell a better model from one that calls more cells parasitised: histogram matching "recovers" 85 to 110% of the MP-IDB drops while destroying specificity on BBBC041. RQ2's share recovered is therefore read together with BBBC041 specificity and AUC for the same step (§5.8).

5.7 Methodological alternatives considered

Most choices follow from one constraint: RQ2 can attribute a change in performance to a step only if the model and the test cells stay fixed in every other way.

Decision	Alternatives	Chosen	Why
Fine-tuning	frozen ImageNet features plus a new head (Rajaraman et al., 2018); all layers trained	all layers trained	stain and cell texture are far from ImageNet photographs, so the early layers need to adapt too
NIH split	random cell-level split (Zhao et al., 2020); patient-grouped split	patient-grouped	a random split puts cells of one slide on both sides; Hou et al. (2026) found accuracy 1.5 points higher per cell than slide-disjoint
Hyperparameter search	grid search or Optuna per model; fixed values from the literature	fixed values	keeps RQ1 a comparison of architectures, not of tuning budgets; fits a free Colab GPU
How to test the explanations (RQ2)	colour augmentation or retraining on target images; editing the test inputs	edit the test inputs	needs no target labels, and the model stays fixed, so each step's effect is attributable; retraining answers a different question
Background removal	rembg U²-Net or IS-Net (Qin et al., 2020); BiRefNet; Otsu on brightness	Otsu on brightness	median Dice 0.90 against the NIH hand-drawn outlines, against 0.03 for U²-Net, which found no cell in 21% of NIH crops; BiRefNet needs 31 s per crop on this laptop
Colour matching	HSV shift; Macenko (Macenko et al., 2009); learned normalisation (Xu et al., 2025); Reinhard and histogram matching	Reinhard and histogram matching	both are simple and fully specified, and the EDA showed they bring the Giemsa stain channels to NIH while an HSV shift moves them away; Xu et al. (2025) found stain-vector methods beat Reinhard on malaria smears, so a Macenko step is a candidate extension

Decision	Alternatives	Chosen	Why
MP-IDB framing	shipped parasite close-ups; whole-cell crops	whole-cell crops	the close-ups cut off the cell edge, and the shipped crops mix masked and unmasked formats by species
Main metric	accuracy, F1 or AUC alone; sensitivity and specificity at 0.5	sensitivity and specificity, with AUC and F1 as secondary	prevalence of 2.7 to 5.1% on BBBC041; the two errors have different clinical costs
Uncertainty	a single test score; cell-level bootstrap; image-level bootstrap	image-level (NIH: patient-level) bootstrap	cells of one image are not independent; gives intervals instead of point estimates

Table 12. Methodological alternatives considered and the choices made.

5.8 Design decisions for the final analysis

The baseline raised eight questions about how the final results will be computed and read. They are settled as follows.

1. **Fixed hyperparameters** (Table 6) for all three arms, so that RQ1 compares architectures and not tuning budgets.
2. **Three seeds per architecture** for the final results; the midterm baseline uses one. The seeds give the run-to-run spread next to the bootstrap intervals.
3. **Paired bootstrap with a Bonferroni correction** for the RQ1 comparison: the 24 comparisons share one 5% error budget, so each interval is widened to 99.8%. At seed 0 the conclusion does not depend on the correction, since 20 of 24 differences stay clear with it.
4. **Share recovered left undefined** unless the raw drop's 95% interval lies above zero.
5. **Miss rate next to the share of misses** in the RQ3 stage table, because the shares follow the stage counts.
6. **The 346 fallback crops are kept**, and RQ3 is also reported without them.
7. **An MP-IDB sensitivity gain counts as recovery only if AUC on BBBC041 does not fall** for the same step, because MP-IDB cannot show whether the model simply calls more cells parasitised.
8. **The AUC drop is reported next to the sensitivity and specificity drops** in RQ1, as a threshold-free view, since the architectures differ mainly in where their scores fall relative to the 0.5 threshold.

6. Reproducibility

All scripts live in the repository and are listed by the pipeline stage they implement (Figure 1).

Stage	Script	Purpose
A	`malaria/paths.py`, `scripts/check_nih_overlap.py`	every dataset root and output folder; NIH-NLM release against cell_images
B	`scripts/build_crops.py` (`malaria/manifests.py`, `crops.py`, `background.py`)	manifests and crops; MP-IDB host-cell framing
C	`malaria/splits.py`	patient-grouped split, pinned in `nih_split.csv`; leakage check
D	`scripts/compute_stats.py`, `compute_colour_stats.py`, `validate_background.py`; `notebooks/03_eda_short.ipynb`	per-cell statistics on tissue pixels; background-method scores; the EDA

Stage	Script	Purpose
E	`scripts/train.py` (`malaria/models.py`, `data.py`); `scripts/pack_for_colab.py`; `notebooks/10_train_colab.ipynb`	fine-tuning on NIH; data archives for Colab; the Colab run
F	`scripts/build_masked_crops.py`, `build_colour_crops.py` (`malaria/colour.py`)	background removal; Reinhard and histogram matching
G	`scripts/evaluate.py`	frozen inference, one prediction file per variant and set, with a `run.json` record
H	`scripts/summarise_results.py`, `compare_models.py` (`malaria/metrics.py`, `results.py`)	RQ1, F1, RQ2 and RQ3 tables per model; paired differences in drop between models
all	`scripts/make_mo_figures.py`, `build_mo_pdf.py`	the figures and the PDF of this document

Table 13. Scripts by pipeline stage. The repository layout is in Appendix D.

To reproduce the work, download the three datasets from the sources in §2.1 into the project root, then run from the repository root:

```
pip install -r requirements.txt
python -m malaria.metrics                       # metric self-test
python scripts/train.py --arch vgg16 --smoke --no-pretrained   # CPU smoke check
python scripts/build_crops.py                   # manifests and crops (A, B, C)
python scripts/build_crops.py --mpidb-wholecell
python scripts/compute_stats.py --workers 6     # EDA statistics (D)
python scripts/compute_stats.py --manifest mpidb_wholecell_cells.csv
python scripts/build_masked_crops.py --method otsu-gray --datasets bbbc041 mpidb_wholecell   # F1
python scripts/build_colour_crops.py            # F2
python scripts/pack_for_colab.py                # code.zip, data_raw.zip, data_rq2.zip
# on Colab, through notebooks/10_train_colab.ipynb, for each of vgg16, resnet50, mobilenet_v2:
python scripts/train.py --arch vgg16 --seed 0 --resume
python scripts/evaluate.py --checkpoint models/vgg16_s0.pt --variants raw masked reinhard histmatch
python scripts/summarise_results.py --model vgg16_s0
# then once:
python scripts/compare_models.py                # RQ1: paired differences in drop
```

The full run order, including the EDA-only steps, is in `README.md`. Seeds are fixed for the split (42), the colour reference sample (0), the bootstrap (0) and each training run (`--seed`).

The code, the pinned split, the manifests' build logic and every summary table are public in the repository. The images are not: anyone reproducing the work downloads the three datasets from the sources in §2.1 and places them in the project root. Training needs a GPU (a free Colab T4 is enough: about three minutes per epoch for VGG-16, under two for the other two models); everything else runs on a CPU. Following Di Cosmo et al. (2026), the repository will be archived with a persistent identifier at the end of the project.

References

Beaulieu-Jones, B. K., & Greene, C. S. (2017). Reproducibility of computational workflows is automated using continuous analysis. *Nature Biotechnology, 35*(4), 342–346. https://doi.org/10.1038/nbt.3780

Di Cosmo, R., Granger, S., Hinsen, K., Jullien, N., Le Berre, D., Louvet, V., Maumet, C., Maurice, C., Monat, R., & Rougier, N. P. (2026). CODE beyond FAIR: A roadmap for reusable research software. *Scientific Data, 13*(1). https://doi.org/10.1038/s41597-026-06705-6

European Parliament and Council of the European Union. (2017). Regulation (EU) 2017/745 on medical devices. *Official Journal of the European Union, L 117*, 1–175. https://eur-lex.europa.eu/eli/reg/2017/745/oj

European Parliament and Council of the European Union. (2024). Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act). *Official Journal of the European Union, L 2024/1689*. https://eur-lex.europa.eu/eli/reg/2024/1689/oj

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 770–778). https://doi.org/10.1109/CVPR.2016.90

Hou, M., He, G., Liu, Z., Huang, J., & Zou, H. (2026). MalariaNet: A microcontroller-deployable malaria-microscopy detector for point-of-care biosensing under leakage-free evaluation. *Biosensors, 16*(7), 358. https://doi.org/10.3390/bios16070358

Hung, J., & Carpenter, A. (2017). Applying Faster R-CNN for object detection on malaria images. In *2017 IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)* (pp. 808–813). https://doi.org/10.1109/CVPRW.2017.112

Kapoor, S., & Narayanan, A. (2023). Leakage and the reproducibility crisis in machine-learning-based science. *Patterns, 4*(9), 100804. https://doi.org/10.1016/j.patter.2023.100804

Kassim, Y. M., Palaniappan, K., Yang, F., Poostchi, M., Palaniappan, N., Maude, R. J., Antani, S., & Jaeger, S. (2021). Clustering-based dual deep learning architecture for detecting red blood cells in malaria diagnostic smears. *IEEE Journal of Biomedical and Health Informatics, 25*(5), 1735–1746. https://doi.org/10.1109/JBHI.2020.3034863

Kingma, D. P., & Ba, J. (2015). Adam: A method for stochastic optimization. *3rd International Conference on Learning Representations (ICLR 2015)*. https://arxiv.org/abs/1412.6980

Ljosa, V., Sokolnicki, K. L., & Carpenter, A. E. (2012). Annotated high-throughput microscopy image sets for validation. *Nature Methods, 9*(7), 637. https://doi.org/10.1038/nmeth.2083

Loddo, A., Di Ruberto, C., Kocher, M., & Prod'Hom, G. (2019). MP-IDB: The Malaria Parasite Image Database for image processing and analysis. In *Processing and Analysis of Biomedical Information* (pp. 57–65). Springer. https://doi.org/10.1007/978-3-030-13835-6_7

Loddo, A., Fadda, C., & Di Ruberto, C. (2022). An empirical evaluation of convolutional networks for malaria diagnosis. *Journal of Imaging, 8*(3), 66. https://doi.org/10.3390/jimaging8030066

Macenko, M., Niethammer, M., Marron, J. S., Borland, D., Woosley, J. T., Guan, X., Schmitt, C., & Thomas, N. E. (2009). A method for normalizing histology slides for quantitative analysis. In *2009 IEEE International Symposium on Biomedical Imaging: From Nano to Macro* (pp. 1107–1110). https://doi.org/10.1109/ISBI.2009.5193250

Otsu, N. (1979). A threshold selection method from gray-level histograms. *IEEE Transactions on Systems, Man, and Cybernetics, 9*(1), 62–66. https://doi.org/10.1109/TSMC.1979.4310076

Qin, X., Zhang, Z., Huang, C., Dehghan, M., Zaiane, O. R., & Jagersand, M. (2020). U2-Net: Going deeper with nested U-structure for salient object detection. *Pattern Recognition, 106*, 107404. https://doi.org/10.1016/j.patcog.2020.107404

Rajaraman, S., Antani, S. K., Poostchi, M., Silamut, K., Hossain, M. A., Maude, R. J., Jaeger, S., & Thoma, G. R. (2018). Pre-trained convolutional neural networks as feature extractors toward improved malaria parasite detection in thin blood smear images. *PeerJ, 6*, e4568. https://doi.org/10.7717/peerj.4568

Rajaraman, S., Jaeger, S., & Antani, S. K. (2019). Performance evaluation of deep neural ensembles toward malaria parasite detection in thin-blood smear images. *PeerJ, 7*, e6977. https://doi.org/10.7717/peerj.6977

Reinhard, E., Ashikhmin, M., Gooch, B., & Shirley, P. (2001). Color transfer between images. *IEEE Computer Graphics and Applications, 21*(4), 34–41. https://doi.org/10.1109/38.946629

Russakovsky, O., Deng, J., Su, H., Krause, J., Satheesh, S., Ma, S., Huang, Z., Karpathy, A., Khosla, A., Bernstein, M., Berg, A. C., & Fei-Fei, L. (2015). ImageNet large scale visual recognition challenge. *International Journal of Computer Vision, 115*(3), 211–252. https://doi.org/10.1007/s11263-015-0816-y

Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., & Chen, L.-C. (2018). MobileNetV2: Inverted residuals and linear bottlenecks. In *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 4510–4520). https://doi.org/10.1109/CVPR.2018.00474

Simonyan, K., & Zisserman, A. (2015). Very deep convolutional networks for large-scale image recognition. *3rd International Conference on Learning Representations (ICLR 2015)*. https://arxiv.org/abs/1409.1556

Xu, T., Wu, Y., Tripathi, A. K., Ippolito, M. M., & Haeffele, B. D. (2025). Adaptive stain normalization for cross-domain medical histology. In *Medical Image Computing and Computer Assisted Intervention, MICCAI 2025* (pp. 24–33). Springer. https://doi.org/10.1007/978-3-032-04981-0_3

Zhao, O. S., Kolluri, N., Anand, A., Chu, N., Bhavaraju, R., Ojha, A., Tiku, S., Nguyen, D., Chen, R., Morales, A., Valliappan, D., Patel, J. P., & Nguyen, K. (2020). Convolutional neural networks to automate the screening of malaria in low-resource countries. *PeerJ, 8*, e9674. https://doi.org/10.7717/peerj.9674

Appendix A. Full RQ2 table

Test set, metric	Model	Raw	Background removed	+ Reinhard	+ Histogram matching
site_a sensitivity	VGG-16	99.3	99.4	80.5	99.8
	ResNet-50	79.3	92.6	42.8	98.4
	MobileNetV2	83.7	84.3	15.9	96.6
site_b sensitivity	VGG-16	89.8	90.8	89.8	100.0
	ResNet-50	68.0	73.6	75.9	95.0
	MobileNetV2	30.4	36.6	47.2	80.5
MP-IDB *P. falciparum* sensitivity	VGG-16	57.4	71.9	66.4	99.5
	ResNet-50	52.6	63.9	52.3	98.1
	MobileNetV2	28.6	27.2	32.9	95.4
MP-IDB other species sensitivity	VGG-16	80.0	82.9	70.0	100.0
	ResNet-50	56.4	67.9	42.1	99.3
	MobileNetV2	38.6	37.9	28.6	88.6

Test set, metric	Model	Raw	Background removed	+ Reinhard	+ Histogram matching
site_a specificity	VGG-16	65.5	53.6	74.9	5.9
	ResNet-50	87.3	66.6	82.0	6.6
	MobileNetV2	90.0	78.9	96.3	44.4
site_b specificity	VGG-16	95.0	86.2	80.3	1.1
	ResNet-50	98.1	98.3	89.8	15.0
	MobileNetV2	99.1	97.2	98.1	41.9
site_a AUC	VGG-16	97.2	96.6	85.6	91.6
	ResNet-50	92.0	90.2	74.5	72.7
	MobileNetV2	94.5	89.6	84.1	88.5
site_b AUC	VGG-16	97.5	95.8	93.7	73.0
	ResNet-50	96.4	96.1	91.6	72.5
	MobileNetV2	93.0	87.2	87.6	69.4

Table A1. Every external set and metric, in %, seed 0. Per-model figures: `R_vgg16_s0.png`, `R_resnet50_s0.png`, `R_mobilenet_v2_s0.png`.

Appendix B. Model information sheet

Field	Value
Task	Binary classification of a single red-cell image as parasitised or uninfected
Input	One RGB cell crop, padded to a square with black, 224 × 224, ImageNet-normalised
Output	A logit; the cell is called parasitised when the probability is at least 0.5
Training population	19,203 NIH cells from 140 patients, one hospital in Chittagong, *P. falciparum* only, about 50% parasitised
External evaluation	BBBC041 site_a and site_b (*P. vivax*, 2.7 to 5.1% parasitised) and MP-IDB (four species, parasitised cells only), without fine-tuning
Intended use	Academic research into why CNN malaria classifiers lose performance across datasets
Out-of-scope use	Diagnosis or screening of patients, or any clinical decision; the model has no clinical validation
Known limitations	Single training site and species; no demographic variables; one seed at the midterm; MP-IDB has no uninfected cells, so specificity is measured on BBBC041 only
Failure modes seen so far	Large loss of sensitivity on MP-IDB *P. falciparum* for every architecture; VGG-16 raises many false alarms on BBBC041 site_a, while ResNet-50 and MobileNetV2 miss more parasites; MobileNetV2 misses most site_b parasites; colour matching shifts scores toward *parasitised* (§5.6)

Appendix C. Continuous-analysis checks

Check	Where
NIH split sizes equal the pinned 19,203 / 4,178 / 4,177; every NIH cell has a split	`data.nih_cells`
No patient in two splits; train and validation patients disjoint	`splits.check_no_patient_leakage`, `train.py`

Check	Where
Every parameter trainable (the weights are unfrozen)	`models.check_all_trainable`
Unlabelled or excluded rows never reach a model	`data.CellDataset`, `data.test_cells`
Reframing keeps every MP-IDB parasite, stage label and cell ID	`build_crops.py --mpidb-wholecell`
Colour-matched manifests keep the row count of their input	`build_colour_crops.py`
Metrics on a perfect predictor: sensitivity, specificity, precision and F1 are 1, drop 0; the weighted AUC and F1 equal scikit-learn's; a set with one class has no precision or F1	`python -m malaria.metrics`
A prediction file is reused only if its checkpoint, manifests, crop contents and row count still match; files are written atomically	`evaluate.py`, `results.load`
Models are compared only if they were scored on the same manifests, crop contents and cells	`compare_models.check_same_inputs`
RQ2 variants hold the same cells	`summarise_results.check_same_cells`
The fallback check removes exactly the 346 MP-IDB cells listed in the framing table	`summarise_results.without_fallback`
The training curves' best epoch and loss match the training log	`make_mo_figures.training_curves`

Appendix D. Repository organisation

```
malaria-cross-dataset/
├── README.md                    # run order and where every crop is saved
├── requirements.txt             # pinned Python packages
├── LiteratureReview_NoahTjonSienKie.pdf   # D2
│
├── malaria/                     # the package every script imports
│   ├── paths.py                 # all dataset roots, output folders and measured constants
│   ├── crops.py                 # Stage B: square_padded_box cropping rule
│   ├── manifests.py             # Stage B: one manifest schema, builders per dataset
│   ├── splits.py                # Stage C: patient-grouped split, leakage check
│   ├── imagestats.py            # Stage D: per-cell features on tissue pixels, separation
│   ├── background.py            # Stages B/F: Otsu masks, host-cell framing, Dice
│   ├── colour.py                # Stage F: Reinhard, histogram matching, stain channels
│   ├── data.py                  # Stage E: nih_cells, test_cells, CellDataset
│   ├── models.py                # Stage E: build_model, check_all_trainable
│   ├── metrics.py               # Stage H: rates, F1, drop, share recovered, bootstrap
│   ├── results.py               # Stage H: reads prediction files back as the reported sets
│   ├── provenance.py            # code commit, manifest hashes and crop digests of every run
│   └── plots.py                 # figure helpers
│
├── scripts/
│   ├── build_crops.py           # Stage B: manifests and crops
│   ├── check_nih_overlap.py     # Stage A: NIH-NLM release vs cell_images
│   ├── compute_stats.py         # Stage D: per-cell statistics
│   ├── compute_colour_stats.py  # Stage D: colour-space statistics
│   ├── validate_background.py   # Stage D: scores background-removal methods
│   ├── build_masked_crops.py    # Stage F1: background removal
│   ├── build_colour_crops.py    # Stage F2: Reinhard and histogram matching
│   ├── pack_for_colab.py        # Stage E: data archives for Colab
│   ├── train.py                 # Stage E: fine-tuning
│   ├── evaluate.py              # Stage G: frozen inference, per-cell predictions
│   ├── summarise_results.py     # Stage H: RQ1/F1/RQ2/RQ3 tables and figure, per model
│   ├── compare_models.py        # Stage H: RQ1 paired differences in drop between models
│   ├── make_mo_figures.py       # figures for this document
│   ├── build_mo_pdf.py          # the PDF of this document
│   └── make_notebook.py         # generates notebooks/01_eda.ipynb
│
├── notebooks/
│   ├── 01_eda.ipynb             # full EDA record (generated)
│   ├── 03_eda_short.ipynb       # short EDA, the version presented
│   └── 10_train_colab.ipynb     # Stages E and G on a Colab GPU
│
├── data/
│   ├── manifests/
│   │   ├── nih_split.csv        # tracked: pins the train/val/hold-out rows
│   │   ├── mpidb_stage_audit.csv  # tracked: source of every MP-IDB stage label
│   │   └── *_cells.csv          # gitignored: rebuilt by build_crops.py
│   └── crops/                   # gitignored: about 87k PNGs, rebuilt by the build scripts
│
├── outputs/
│   ├── figures/                 # EDA and results figures
│   ├── tables/                  # summary tables (rq1/rq2/rq3, F1, colour reference, overlap)
│   └── predictions/             # gitignored: per-cell predictions (kept on Drive)
│
├── models/                      # gitignored: trained weights (0.5 GB per VGG-16)
│
└── docs/
    ├── methodology_overview_v3.md   # this document (D3), and its PDF
    ├── methodology_overview.md  # the earlier Markdown version
    ├── elsa_report.md           # D5: final checklist, mitigations, panel reflections
    ├── elsa_checklist.md        # the earlier working checklist
    └── methodology/figures/     # figures used in this document
```

The four source datasets (`cell_images_NIH/`, `malaria_BBBC041/`, `MP-IDB-…/`, `NIH-NLM-ThinBloodSmearsPf/`) sit in the project root and are gitignored. Images and crops are never committed, because of their size and because BBBC041's licence requires derived data to be shared under the same terms.